

Text Comprehension: Heads-Up vs. Auditory Displays

Implications for a Productive Work Environment in SAE Level 3 Automated Vehicles

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Figure 1: Driving simulator setup in the “Heads-Up” condition during automated driving (left) and when a Take-Over Request is issued (right).

ABSTRACT

With increasing automation, vehicles could soon become “mobile offices” but traditional user interfaces (UIs) for office work are not optimized for this domain. We hypothesize that productive work will only be feasible in SAE level 3 automated vehicles if UIs are adapted to (A) the operational design domain, and (B) driver-workers’ capabilities. Consequently, we studied adapted interfaces for a typical office task (text-comprehension) by varying display modality (heads-up reading vs. auditory listening), as well as UI behavior in

conjunction with take-over situations (attention-awareness vs. no attention-awareness). Self-ratings, physiological indicators, and objective performance measures in a driving simulator study ($N = 32$) allowed to derive implications for a mobile workspace automated vehicle. Results highlight that heads-up displays promote sequential multi-tasking and thereby reduce workload and improve productivity in comparison to auditory displays, which were still more attractive to users. Attention-awareness led to reduced stress but later driving reactions, consequently requiring further investigations.

CCS CONCEPTS

• **Human-centered computing** → Empirical studies in HCI; *Interactive systems and tools*; • **Applied computing** → Computers in other domains.

KEYWORDS

automated driving, take-over requests, multitasking, productivity, mobile office, displays, attention-awareness

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1 INTRODUCTION

At least since the alphabetization of the masses and the “reading revolution” in the 18th century [12], an inevitable consequence following the invention of the printing press by Johannes Gutenberg, reading became an integral part of our daily lives, accelerating the speed of civilization. Whereas the downfall of reading has been proposed various times, it survived the ages, always adopting to new technologies. Today, people seem to be reading or listening (i.e., *comprehending text*) even more than ever before [1]. Thus, it is not surprising that 52.7% of a web survey’s respondents claimed reading to be desired in automated vehicles (AVs) [44]. Comprehending text (whether it be via reading or listening) is integral to many jobs, especially in typical office work, where emails, reports and forms of written communication is substantial. *Doing office work on the motorway* was part of the daily lives of many professions, including researchers, as early as in 1998 [29]. Commuters nowadays seem to engage in work-related activities frequently to compensate for time losses [43]. Both businesses and consumers could benefit from transforming AVs into “mobile offices” [10], allowing to improve work-life balance by a more useful utilization of commuting and travel time and thereby increasing productivity but also leisure time. In addition, this could even allow maintaining jobs that potentially suffer from vehicle automation. For example, truck drivers may, in the future, perform additional activities, such as logistics. The importance of these topics has also recently caught the attention of AutomotiveUI researchers, who started to extend the term of Non-Driving Related Tasks (NDRTs), that was traditionally mostly connected with entertainment and informal communication, with office work, and, accordingly, engaged in discussions about opportunities and requirements [10, 49].

However, quick realization seems unlikely: As a result of recent incidents with already available automated driving systems operating on SAE level 2 [55], it has been stated that driving automation frequently requiring human intervention is unsafe [56], and some vehicle manufacturers claim to even skip SAE level 3 “conditional automation” and directly move towards higher levels of automation [61]. Thus, it is the burden of HCI/human factors research to develop

technology supporting level 3, demonstrating that conditionally automated driving, allowing drivers to switch between side activities and the driving task, is safely possible. We believe this can be achieved by integrated support of NDRTs with properly designed human-machine interfaces (HMIs). With these issues in mind, this work addresses the question of how SAE level 3 AVs can provide effective interfaces for office work. They should allow both comfortably and productively engaging in NDRTs, as well as safely responding to automation-initiated control transitions (so-called “Take-Over Requests”, TORs).

Distraction versus Engagement

Research in manual driving or automation below level 3 consistently reports that secondary task engagement is detrimental to driving performance [42, 69]. Legislation in most countries prohibits those activities, including the usage of nomadic consumer devices, for that very reason. In contrast to manual driving, however, NDRT-studies in SAE level 3 automated driving are reporting that NDRT engagement does not necessarily have negative effects on driving performance in TOR situations, but results of these studies are not very clear-cut [14, 39, 54]. The Malleable Resources Theory by Young and Stanton [70] even postulates that engaging in NDRTs may be necessary, as the availability of cognitive resources adapts to the level of workload. In turn, long phases of passiveness (i.e., when driving without a specific task besides *staying take-over ready* in a conditionally AV) make it hard to respond to cognitively demanding situations (such as an imminent TOR), as prolonged driving automation can induce drowsiness [28]. Thus, **NDRTs should be available but calibrated** in their cognitive demand [70], such that “distraction becomes engagement” [38]. We hypothesize **this may only be possible consistently with the introduction of new user interfaces (UIs) or adoption of existing UIs** to the domain of SAE level 3 automated driving since especially the utilization of handheld devices significantly degraded TOR reaction time and post-TOR driving control in many previous experiments [37]. Further restrictions (e.g., by law) seem unfeasible as an alternative solution, considering the potential co-existence of various levels of automation and people’s desire to increasingly engage in NDRTs with rising automation levels [32, 45]. Hence, automotive human-computer interaction researchers need to provide suitable, domain-aware, interfaces.

Optimizing NDRT Interfaces for SAE Level 3

Two promising emerging possibilities for adoption are: UI 1) modality and 2) behavior. In this work, we focus on the display modality and UI behavior of the NDRT under investigation: reading/text comprehension in a productive work setting. We identified two promising display modalities that

have already found their way into production vehicles for mostly driving-related information and compared their effectiveness in the “mobile office” context: Heads-Up (HUDs) and Auditory Speech Displays (ASDs). Considering behavior, we investigate the behavioral aspect of “attention-aware user interfaces”, that recently have demonstrated their beneficial impact on sequential multitasking in vehicles [67, 68]. Both aspects (display modality and attention-awareness) were examined in a driving simulator study (N=32). Regarding evaluation, we based our approach on the combination of ISO 9241-11 for usable (i.e., effective, efficient and satisfactory; [25]) interfaces and SAE J3114 for human factors for automated driving systems [17]. We, thus, triangulate behavioral (TOR-, NDRT performance, eye-tracking), physiological (galvanic skin response, pupil dilation), and self-rating (self-rated workload, acceptance, and usability) measures. Using this, we are able to give broadly applicable recommendations regarding the implementation of productivity-oriented NDRT-UIs in SAE level 3 AVs. While we base the NDRT-UI’s attentiveness on previous work [67, 68], we compare HUDs vs. ASDs for productivity-oriented NDRTs in level 3 vehicles, as a first, to the best of our knowledge.

2 RELATED WORK

In the domain of conditionally automated driving a plethora of research examining take-over situations, and the influence of NDRTs on them exists, but works on productivity-oriented tasks are scarce, especially in comparison to typical infotainment interaction (see [37] for an exhaustive review on related studies).

UI Display Modalities - Visual vs. Auditory

Using standardized and naturalistic tasks, researchers report driving performance advantages when using auditory-vocal in comparison to visual-manual tasks [62]. Wicken’s Multiple Resource Theory [64] predicted these results by stating that two tasks with the same modalities will induce greater interference than with different modalities. Thus, in the visual-manual driving task, auditory-vocal UIs seem to be more promising than visual displays, and many options, especially for driving-related information, have been studied extensively [26, 46], and also applied in actual vehicles [2]. Still, Chang et al. [9] report that especially interactively talking (i.e., not just listening) has a significant impact on cognitive workload, which may influence driving performance. Another arguable downside of (simple) ASDs, especially considering office work settings where productivity is important, is the inability to employ efficient perception patterns (like skipping unimportant words). Further, text-to-speech systems’ quality can also significantly influence comprehensibility [24] – problems visual displays

do not have, as readers can adjust their reading speed and patterns dynamically, and re-read accordingly, if needed. HUDs, additionally, can also reduce the detrimental effects of traditional visual displays in the center stack [23, 31, 60]. We have confirmed these results in our previous experiments [51, 52], where we investigated HUDs vs. heads-down displays or brought-in devices in the context of productive text entry during conditionally automated driving. The Cognitive Control Hypothesis by Engström et al. [18] also argues that (driving) performance may not be undermined despite occupying the same resource, if the task is automatized (i.e., well-practiced, consistently mapped). Users may perform better with HUDs as the visual road-observation task may be barely conscious. However, technical limitations of HUDs have to be considered (e.g., with sun glare [20]). A direct comparison of visual and auditory modalities by Weinberg et al. [63], in a list-selection task in manual driving, showed that the ASD still had the least detrimental effects on driving performance but significantly reduced task performance (i.e., productivity). A study by Louw and Merat [33] investigated gaze behavior during automated driving and found that “scenarios that encourage driver gaze towards the road are more likely to increase situation awareness”. Accordingly, a HUD could direct gaze towards the road, but overlaying the windshield can also degrade driving performance [58], cause a loss of global situation awareness [19] and cause change/inattention blindness [22]. In summary, the display modality of choice for a “mobile office” AV is not obvious as it remains mostly unexplored in the domain of productive NDRTs, while results are not clear-cut in others, and thereby requiring further investigation.

UI Behavior - Attention Awareness

In previous research addressing TORs, we have proposed the provision of “attentive user interfaces” (AUIs) in AVs. AUIs are “UIs that are sensitive to the user’s attention”, and “structure communication such that the limited resource of attention is allocated optimally across the user’s tasks” [59, 68]. The underlying idea is, that users should be interrupted only in “opportune moments” [48], for example at times of low mental workload, or at the presence of task boundaries [3], to reduce the negative effects of so-called “interruption” and “resumption lags”, which describe the time and effort needed to rehearse and recall the state of ongoing tasks when sequentially switching between activities [47]. We have investigated these issues in a driving simulator study addressing text messenger input on handheld devices, and could thereby demonstrate the positive effects of AUIs (less stress and higher performance) in scheduled TOR situations by the integration of handheld devices into the vehicle’s ecosystem [67]. We believe the approach is valuable for our current problem domain too – experiments investigating the issues

of sequential multitasking have been conducted in office settings previously [3], and in this context, users' attention and workload is a sensitive resource that should not be trespassed. Frequently interrupting drivers in work-related tasks could potentially hinder productivity, increase workload and stress, and thus degrade safety. Consequently, we want to investigate if issuing TORs after, rather than during subtasks, also benefits integrated HMIs for text comprehension. While we addressed scheduled TORs with a relatively long lead time (LT) of 12 seconds in former experiments [50, 67, 68], we now investigate if the concept also improves imminent transitions during engagement in productivity-oriented NDRTs. According to SAE J3114 [41], scheduled TORs represent expected transitions, while imminent TORs are unexpected, for example due to emergency situations. Some could question if it is possible to delay transitions in emergencies, however, in our proposed system architecture for AUIs in AVs, we have demonstrated how future vehicle-to-vehicle (C2X) communication systems can transform imminent into scheduled transitions at least for following vehicles [68]. Strictly speaking, such transitions can then again be called "scheduled", but imply shorter LTs. Thus, in this experiment we investigate the potential of AUIs in TORs with only 5 sec. LT.

3 METHOD AND HYPOTHESES

Following these considerations, we investigated display modality and attention-awareness for productivity-oriented reading NDRT-UIs in the domain of conditionally automated driving (not manipulated was text difficulty). Subsequently presented are the research questions and respective hypotheses, followed by a description of the prototype design and user study executed.

RQ1 - NDRT Display: How do the domain-adapted display modalities for text-comprehension impact TOR responses, workload levels, and productivity differently?

H1 - TOR Responses The auditory speech display leads to significantly better TOR responses than the heads-up display. *We believe this will be the case as drivers will be able to continuously monitor the road environment.*

H2 - Workload Self-ratings contradict physiologically measured workload. *Since drivers do not have to perform two concurrent visual tasks, they believe the auditory speech display to be less demanding, yet the additional second modality (inclusion of an auditory task) may increase workload.*

H2.1 The auditory speech display leads to significantly lower self-rated workload than the heads-up display.

H2.2 The auditory speech display leads to significantly higher physiologically measured workload than the heads-up display.

H3 - Productivity Self-rated acceptance and usability contradict measured productivity. *Reading with the auditory speech display allows to solely lead visual resources to the driving environment, yet we believe visual reading can achieve higher NDRT performance, as participants will be able to adjust their reading behavior to the visually presented information.*

H3.1 The auditory speech display leads to significantly higher acceptance and usability ratings than the heads-up display.

H3.2 The heads-up display leads to significantly higher measured productivity (i.e., speed and correctness) than the auditory speech display.

Additionally, we want to investigate attention-aware TORs (scheduling them to be issued after rather than during subtasks, based on [67, 68]), and adopt related hypotheses hypotheses accordingly.

RQ2 - NDRT Attentiveness: Does scheduling TORs at subtask boundaries instead of during task execution have significant beneficial impact on TOR performance (quantified as reaction times and driving performance) and (physiologically measured) stress/arousal levels in TOR situations, while engaged in a reading-comprehension task with adopted interfaces during automated driving phases?

Prototype Implementation

As a **HUD**, we showed a semi-transparent (50%) white window with black font on top of the driving simulation's visualization, centered on the top third and in the left two thirds of the windshield (cf. Figure 2). These decisions are based on recommendations by Tsimhoni et al. [58] to not center the HUD. Additionally, this approach allows us to compare pupil dilation to the auditory speech display, as the focal plane remains the simulator's visualization and lighting is constant.

The potential of freeing the text comprehension task from requiring visual attention by using an **Auditory Speech Display (ASD)**, seems obvious [46] – but how ASDs' demands (i.e., workload requirements) affect drivers in conditionally automated driving in comparison with the HUD, is questioned for the first time in this work (to the best of our knowledge). We utilized the free Google Text-To-Speech Engine for speech synthesis with the default female voice as the ASD. Participants were required to set their talking-speed preference from 0.5x-3.0x the engine's default speed (based on repeatedly playing an example sentence in the 7 available speed levels) before using the interface.

In order to be able to evaluate RQ2, the NDRT state (started subtask, finished subtask) was consistently reported to the simulated automated driving system.

4 DRIVING SIMULATOR STUDY

To evaluate the hypotheses in a conditionally AV, we designed and executed a within-subjects study in a hexapod driving simulator. The driving simulator allowed us to test safety-critical TOR situations while still maintaining a high degree of fidelity. In order to be able to validate the hypotheses, we need two **independent variables (IVs)** and respectively two levels for each of them:

NDRT Display (D) :

Heads-Up Display (HUD) vs. Auditory Speech Display (ASD)

NDRT Attentiveness (A) :

TORs *after* vs. *during* subtasks

IV A was hidden in the display modality conditions, in the form of alternating TORs *after* or *during* subtasks (i.e., after or during the display of a sentence). All tools involved in the study (measurements devices, environment simulation, prototypes) were synchronized and execution automated by in-house software [53].

Procedure

After receiving written and verbal explanations of the upcoming experiment, each participant signed a consent form on legal, data privacy, and safety aspects. Then, they underwent a 45-minute phase in which they filled various questionnaires on the topic of automated driving, which we will discuss in a different publication. Subsequently, they were forwarded into the driving simulator cockpit, where they were equipped with various physiological measurement devices (we focus on Galvanic Skin Response (GSR), and eye-tracking in this report) and received instructions on how to perform the upcoming dual-task (conditionally automated driving plus NDRT). Before the actual test started, they underwent a 5 minutes-long training drive, in which they had to accomplish two simplified take-overs with no NDRT. Then, they proceeded with the test drives (ASD and HUD respectively – IV A was hidden in those two drives), whereas conditions were counterbalanced throughout participants. Before each drive, the eye-tracker was calibrated, driving simulation visualization initialized, and the first 20 seconds used as the baseline for normalized pupil dilation measurement (doors closed, frozen driving simulation visualization, with HUD in respective condition). After successfully executing each drive (about 12 minutes each), participants were instructed to fill a set of self-rating scales (“Raw” NASA TLX [21] and the Technology Acceptance Model [13, 30]) based on their “holistic experience of automated driving, performing the NDRT, and the Take-Overs”. Subsequent to finishing both drives, they underwent a debriefing and received their participation compensation.

Dual-Task: Reading/Listening and Take-Overs

In each test condition, participants started driving in the middle lane of a straight 3-lane German-layout highway with 100 km/h in automated driving mode (AD). In this phase, they had to accomplish the reading-comprehension task *as fast and as accurate as possible*, but also *stay take-over ready*. Three lead vehicles (one on each lane) were present at the same time-distance (5 seconds) at all times – which thereby also represented the TOR LT. This phase ended/was interrupted by a TOR signal that caused two of the three front vehicles (middle vehicle, plus either left or right) to suddenly stop. The TOR was either issued *during* a subtask, (i.e., after a subtask was visible for 1.5 seconds) but before it was finished (i.e., a task-response issued by the user), or *after* a subtask / user’s task response / timeout (IV A) in an alternating fashion (i.e., subsequent to each TOR *during* a subtask, a TOR *after* followed and vice versa). In response to each of the four TORs per drive, the *driver-worker* had to take-over control by actuating the driving controls (cf. TOR measurement recommendations [66]: $> 2^\circ$ steering wheel angle change or $> 5\%$ gas/brake pedal actuation) in order to perform a lane change to the only free lane, avoid the obstacles, and ultimately hand back control to automation by re-aligning in the middle lane and keeping the lateral position (± 0.5 meters) for 2.5 seconds without interruption. Automation state was always conveyed visually (cf. Figure 2 (2); green: automated mode, red: TOR issued, gray: manual driving (MD)) and important state transitions also accompanied acoustically with a corresponding auditory signal sound (AD $\rightarrow$ TOR, MD $\rightarrow$ AD; cf. Figure 1). This dual-task procedure was repeated four times per condition: two take-overs *during* subtasks in the trial, two *after* (alternating timing); the free lane (left or right) and vehicle appearances were randomized. The NDRT was paused during the TOR/MD phase and resumed after the handback (i.e., returning to AD) with a new subtask, i.e., a new sentence given to the user (the old sentence was discarded and also not included in the analysis).

Reading/Text-Comprehension Task. We based the reading-comprehension task on the reading-span task by Daneman and Carpenter [11], which is frequently utilized for cognitive processing and reading comprehension experiments [65]. Participants in the presented experiment had to perceive (i.e., via the HUD or ASD) single sentences, then process each of these “subtasks”, in order to directly give binary responses on their semantic correctness. Time available from issuing each sentence (reveal on the HUD or start of audio playback) until a participant’s binary response was limited to 10 seconds, otherwise, a timeout occurred. Upon each response, participants received feedback about whether they were correct, incorrect or too slow to respond, and subsequently the next subtask. This feedback was given via the



Figure 2: Driving Simulator Cockpit from participants' perspective (wide-angle view), with (1) GSR electrodes attached to the left hand's middle and ring finger, (2) observation camera, (3) automation-state feedback and (4) the computer mouse for task-responses. The HUD is highlighted with a red border.

same modality as the respective condition (i.e., in condition HUD: green checkmark, red "X", or an illustration of a snail; in condition ASD: spoken "correct", "incorrect", "too slow" respectively). Responses were issued by the user via a computer mouse, which had the left mouse button highlighted in green for "semantically correct", and red for "semantically incorrect". The mouse was positioned and fixed in the driving simulator cockpit's center console, cf. Figure 2 (4). This input device was chosen as we assumed users' familiarity with computer mice and thereby low requirements of visual attention, similar to traditional haptic buttons in the center console. Instructed to be "as fast, but also as correct, as possible", participants rated about 80 sentences during each condition. All sentences were in German, with participants being required to be either native Germans or have high language proficiency.

Measures

In this study we employ a variety of self-rating scales, as well as physiological and performance measures, as listed in Table 1 – for details, we refer to the cited articles. To measure physiological responses we utilized a professional 500 Hz GSR measurement system by G.TEC MEDICAL ENGINEERING GMBH (gtec.at), and for eye-tracking the head-mounted Tobii Pro Glasses 2 with 50 Hz. For GSR application we followed recommendations by [7, 8] and analyzed this indicator for arousal in the sympathetic nervous system regarding event-related (phasic) Skin Conductance Responses (SCRs) with the open-source Matlab toolbox Ledalab [4] and its Continuous Decomposition Analysis method [5, 6] for 10 seconds after each TOR (GSR responses are delayed by about 1 second and approx. 11 seconds was the minimum time to handback to

Table 1: Analyzed self-rating scales, physiological, and performance measures.

Measure	Indicators
<i>Pre-test</i>	
Demographics	Age, Gender, Mileage, Occupation
<i>During whole condition, measured for each completed drive</i>	
NDRT Perf. (Productivity)	Average Task Response Time (RTt), F1-Score [15, 57]
Workload (physio. measured)	(Tonic) Skin Conductance Level (SCL, [8, 40]), Normalized Pupil Dilation [35]
<i>TOR Responses, measured for each TOR/manual driving phase, 4 times per drive</i>	
Take-Over Reaction (Safety I)	Time to First Driving Action (RTd), Time to First Gaze on Road (RTe)
Driving Performance (Safety II)	Average Brake Actuation (Brake Act.), Time-To-Collision on Lane Change (TTCoLC, [16]), Steering Wheel Reversal Rate (SRR, [36])
Stress / Arousal	(Phasic) Skin Conductance Responses (SCRs, [5, 6, 8])
<i>After each condition</i>	
Workload (Self-Ratings)	"Raw" NASA Task Load Index (NASA TLX, [21])
Usability, Acceptance	Technology Acceptance Model (TAM, [13])

automated driving over all TORs in this study). As a measure of tonic activity (i.e., longer-term effects of workload), we averaged the Skin Conductance Level (SCL, [40]) over each condition. For a differentiation of GSR regarding SCL and SCRs, we refer to [34]. To analyze pupil dilation as another measure of workload [35], we utilized a custom Matlab pipeline with Kret and Sjak-Shie's open-source pre-processing toolbox [27]. Driving performance data was fetched from the driving simulation (IPG CarMaker), and task performance from our respective prototype applications.

Participants

In total, 32 participants (17 female, 15 male) between 20 and 70 years (women: $M = 31.82$, $SD = 14.94$, Min. = 20, Max. = 70; men: $M = 33$, $SD = 15.02$, Min. = 21, Max. = 64) took part in this study. All of them had a valid driver's license, normal or corrected-to-normal vision, were fluent in German, i.e. the language in which the reading comprehension task was provided, confirmed to not have consumed any alcohol or drugs, and reported no diagnosis of a psychiatric disorder. The average self-estimated mileage was 12781.25 km per year with a standard deviation of 9816.82 km (Min. = 1000, Max. = 40000).

Sampling Procedure

The study was approved by the ethics committee of the Catholic University of Eichstätt and conducted in accordance with the General Data Protection Regulation of the European Union. All participants gave their written informed consent prior to participation in the study. Participants received 20 Euro as compensation, except employees of the local institution, who could participate in their working hours but could not receive monetary compensation due to legal reasons.

Statistical Data Analysis

We applied paired two samples *t*-tests, and Wilcoxon signed-rank tests where applicable, for dependent variables (DVs) measured per-drive level (cf. Table 1) – this includes post-condition self-ratings. A detailed analysis of the affected DVs on TOR-level will be subject to a future publication. For measures analyzed on TOR-level (i.e., 4 times per drive, cf. Table 1), we applied a two-way repeated measures ANOVA. The Shapiro Wilk's tests revealed that normality could be assumed in almost all conditions. For the three non-normal conditions, no data transformation yielded normally distributed data. Hence, we inspected the QQ-plots for these few non-normal conditions, which indicated no serious deviations from normality. Besides this, we sampled $N = 32$ persons, to exceed the central limit theorem ($N \geq 30$), to stabilize any statistical results of our study. In case of multiple testing yielding to multiple comparisons, we adjusted the overall significance level $\alpha = .05$ to the respective nominal (Bonferroni-corrected) significance level α' to not inflate the Type I Error rate.

5 RESULTS

We conducted three main statistical analyses. A description of DVs can be found in section Measures and the corresponding Table 1. Eye-tracking data of 4 participants had to be excluded due to technical issues (corrupted recordings). In condition ASD, 4 participants chose the text-to-speech engine's default speed (1x), 23 chose 1.5x and 5 chose 2x. Other speeds (0.5x, 2.5x, 3x) were not selected.

TOR Response

First, we performed a two-way repeated measures ANOVA with the two IVs D and A. Descriptive statistics are listed in Table 2 and ANOVA results with test statistics, significance values, and effect sizes in Table 3.

We found a medium effect for take-over reaction RTd. In both conditions (HUD and ASD), participants responded quicker, yielding a shorter reaction time, if a TOR was issued during a subtask, compared to when a TOR was issued after the completion of a subtask. Additionally, we could uncover a medium effect for GSR after TORs for SCR and for Phasic

Maximum. Both measures revealed higher values if a TOR was issued *during* the reading comprehension task for the HUD as well as the ASD, which is associated with higher amounts of mental stress. However, we could identify no significant effect for the outcome measures of time to first gaze on road (RTe), driving performance after TORs (TTCOLC, SRR, and Brake Act.), and of the GSR parameter nSCR. Moreover, no interaction effects could be identified in any of the combinations of the conditions tested on the DVs.

NDRT Performance and Workload

Second, we applied a paired two samples *t*-tests for NDRT performance and physiologically measured workload to compare the conditions of IV A (HUD vs. ASD). The statistical analyses (cf. Table 4 for descriptive and test statistics) showed a substantially better average task response time (RTt) for the HUD than for the ASD. Additionally, the measurements of the pupil diameter are in line with the task response time. Hence, the average normalized pupil diameter was substantially smaller for the HUD than for the ASD, indicating lower cognitive workload. However, no advantage of either the HUD or the ASD could be identified for the DVs F1-Score and SCL.

NASA TLX and TAM

For the statistical analyses of the (ordinal) self-rating scales, i.e., the NASA TLX and the questionnaire for the Technology Acceptance Model (TAM), we applied the non-parametric Wilcoxon signed-rank test. Descriptive (medians) and test statistics including effect sizes can be found in Table 5.

In advance to performing the tests, we performed reliability analyses and computed Cronbach's Alpha (CA) for each dimension (except, the single-item dimension "intention to use the system (Int)" of the TAM; Note: it is not possible to compute CA for a single-item dimension), which resulted in satisfying reliability scores for the sum score of the NASA TLX ($CA_{(HUD)} = .78$; $CA_{(ASD)} = .76$) and for all three dimensions of the TAM (perceived usefulness (PU): $CA_{(HUD)} = .87$; $CA_{(ASD)} = .93$; perceived ease of use (PEOU): $CA_{(HUD)} = .83$; $CA_{(ASD)} = .81$; attitude towards using the system (ATT): $CA_{(HUD)} = .91$; $CA_{(ASD)} = .82$).

The Wilcoxon signed-rank test showed a sufficiently large effect in favor of the ASD in regards to the sum score of the NASA TLX, as well as for its dimensions mental demand, physical demand, and effort. For the dimension ATT of the TAM also a large effect emerged in favor of the ASD. However, no statistically meaningful differences were found for the other NASA TLX dimensions temporal demand, performance, and frustration, as well as the remaining dimensions of the TAM (PU, PEOU, and Int; cf. Table 5).

Table 2: Descriptive statistics: mean (M) and standard deviation (SD) for TOR reaction, driving performance during manual driving, and GSR results within 10 seconds after a TOR.

	Heads-Up		Auditory Speech	
	After	During	After	During
	M (SD)	M (SD)	M (SD)	M (SD)
Take-Over Reaction				
RTd (sec.)	2.61 (0.67)	2.55 (0.56)	2.85 (0.62)	2.55 (0.53)
RTe (sec.)	0.189 (0.220)	0.217 (0.216)	0.273 (0.868)	0.079 (0.139)
Driving Performance during manual driving				
TTCOLC (sec.)	3.07 (1.37)	3.23 (0.66)	3.19 (0.86)	3.33 (0.79)
SRR (r/min)	61.46 (11.79)	63.59 (13.51)	58.96 (13.43)	62.18 (13.93)
Brake Act. (norm.)	0.028 (0.038)	0.031 (0.035)	0.027 (0.033)	0.039 (0.044)
Galvanic Skin Response after TORs				
SCR (μS)	1.70 (0.96)	1.89 (1.18)	1.68 (1.13)	1.96 (1.13)
nSCR (n)	7.56 (1.96)	7.19 (2.29)	6.91 (2.33)	7.06 (2.52)
Ph.Max (μS)	5.55 (3.29)	6.54 (3.68)	5.38 (2.73)	6.54 (3.41)

Table 3: Two-way repeated measures ANOVA statistics: TOR Reaction, Driving Performance during manual driving, and (phasic) GSR results within 10 seconds after a TOR.

	HUD vs. ASD		After vs. During		Interaction	
	F	p (η_p^2)	F	p (η_p^2)	F	p (η_p^2)
Take-Over Reaction ($df(RTd) = 1, 31$; $df(RTe) = 1, 27$)						
RTd	2.60	.117 (.08)	6.92	.013 (.18)*	4.19	.049 (.12)
RTe	0.105	.749 (.00)	0.891	.354 (.03)	1.81	.189 (.06)
Driving Performance during manual driving ($df = 1, 31$)						
TTCOLC	0.341	.564 (.01)	1.77	.194 (.05)	0.012	.91 (.00)
SRR	1.30	.263 (.04)	2.30	.139 (.07)	0.065	.80 (.00)
Brake Act.	0.42	.521 (.01)	5.89	.021 (.16)	2.83	.103 (.08)
Galvanic Skin Response after TORs ($df = 1, 31$)						
SCR	0.021	.885 (.00)	6.92	.013 (.18)*	0.27	.604 (.00)
nSCR	0.757	.391 (.02)	0.368	.548 (.01)	1.72	.199 (.05)
Ph.Max	0.040	.842 (.00)	10.81	.003 (.26)*	0.056	.815 (.00)

* ... significant at the nominal (Bonferroni-corrected) significance level $\alpha' = .025$ (Take-Over Reaction) and $\alpha' = .0167$ (Driving Performance and GSR after TORs); Interaction: (HUD vs. ASD) x (After vs. During)

Table 4: Descriptive statistics: means (M) and standard deviation (SD); and paired two samples *t*-test statistics for NDRT performance (productivity) and physiologically measured workload.

	Variable	Heads-Up	Auditory Speech	<i>df</i>	<i>t</i>	<i>p</i>	Cohen's <i>d</i>
		M (SD)	M (SD)				
NDRT Perf. (Productivity)	RTt (sec.)	4.87 (1.07)	5.33 (0.58)	31	-2.57	.015*	.45
	F1-Score	0.90 (0.07)	0.89 (0.06)	31	-3.76	.722	.14
Workload (physio. measured)	SCL (μ S)	9.78 (5.08)	10.51 (5.88)	31	-2.03	.051	.32
	Pupil diam. (norm.)	0.98 (0.04)	1.02 (0.07)	27	-3.24	.003*	.67

* ... significant at the nominal (Bonferroni-corrected) significance level $\alpha' = .025$

Table 5: Descriptive statistics: median (Mdn); and Wilcoxon signed-rank test statistics for NASA TLX scores (6x 10-point rating scales, 1–10), and TAM dimensions (10-point rating scales, 0–9).

		Heads-Up	Auditory Speech			
	Dimension	Mdn	Mdn	Z	p	Effect size
NASA TLX	Sum	27	24	-3.76	.000*	.83
	Mental Demand	6	4	-3.52	.000*	.74
	Physical Demand	4	2.5	-2.86	.004*	.54
	Temporal Demand	4	3	-2.54	.011	.51
	Performance	7	7	-0.93	.354	.17
	Effort	4	3.5	-3.23	.001*	.70
	Frustration	3	2	-0.85	.393	.18
TAM	Perceived Usefulness	12	12	-1.31	.192	.19
	Perceived Ease of Use	14	14	- 0.77	.443	.15
	Attitude towards the system	12	13	-2.67	.008*	.44
	Intention to Use the system	4	4	-2.23	.026	.39

* ... significant at $\alpha = .05$ significance level for the overall sum score of NASA TLX, or at the nominal (Bonferroni-corrected) significance level $\alpha' = .0083$ for the six dimensions of NASA TLX, and at $\alpha' = .0125$ for TAM.

6 DISCUSSION

In the following, we discuss the the results with respect to the postulated research questions and hypotheses, which reveals an interesting picture.

RQ1 – NDRT Display: HUD vs. ASD

First, we wanted to investigate the impact of NDRT feedback modality on safety, i.e., initial gaze and driving response to TORs, and driving performance until the handback (RQ1, H1). Interestingly, we could not find a significant difference in driving performance in our setting, leading us to **reject H1**. This could hint at the fact that placing the visual display on the windshield, instead of the center stack or similar, is similarly effective as shifting to auditory feedback.

Comparing self-rated workload to physiologically measured workload indicators (skin conductance level and pupil dilation) indicate that self-rated workload (NASA TLX) is higher in the HUD condition than in condition ASD, letting us **accept H2.1**. In contrast, physiological indicators of tonic workload conclude in favor of the HUD, as the baseline-corrected pupil diameter was significantly lower (i.e., lower workload) than in the ASD condition. Although Cohen's d would reveal a small effect ($d = .32$) for SCL too, we could not uncover a statistically meaningful difference here. In total, we argue for **accepting H2.2, and accordingly H2**. To further investigate this matter, we additionally analyzed horizontal gaze dispersion (standard deviation of gaze position, [33]) and found a significantly higher gaze dispersion when using the auditory speech display, compared to the HUD (two-sided paired-samples t -test; $F(1, 27) = 4.039, p < .001$, Cohen's $d = 0.76$; $M_{HUD} = 9.02, SD_{HUD} = 2.62, M_{ASD} = 11.95, SD_{ASD} = 3.07$). This shows that, on average, participants

were more focusing on the windshield with a HUD, or, looked around more with the ASD. Being able to “look around” could suggest participants to be more free in observing what they want and thus lead to lower self-rated workload and higher attractiveness. Ultimately though, it induced more workload as indicated by higher activity of the sympathetic nervous system, as participants actively perceived the environment in their field of view, in parallel to the auditory task, instead of just focusing on the NDRT or sequentially switching between the NDRT and driving task. However, while the level of cognitive demand is measurably higher with the ASD, this freedom may allow for better allocation of situation awareness too, which should be subject of further investigations.

Considering productivity as an indicator of usability, hypothesis H3 is in line with H2 by stating that the ASD is better than the HUD in terms of self-ratings, but measurably performs worse. The analysis, however, only partly confirms this hypothesis. Users had a significantly higher Attitude towards the ASD, but other TAM dimensions were not substantially different. These results only let us **partly accept H3.1**. Regarding productivity, we found a significant effect for average task response time, i.e., the speed with which the NDRT was completed was higher in the HUD condition while the accuracy (F1-Score) was not significantly different. This again leads us to **partly accepting H3.2**. One relevant factor regarding response time is the fixed amount of time needed for listening, i.e., the text-to-speech engine needs a minimum time to output the whole sentence, depending on the pre-configured output-speed. Visual reading speed and patterns, on the other hand, could be adjusted on demand. In summary, we **partly accept H3** in regards to acceptance ratings and measured task completion speed.

RQ2 – NDRT Attentiveness

Considering the obtained phasic GSR results, we found a positive effect of AUIs on stress for TORs with a 5 seconds LT and NDRTs performed on in-vehicle HMIs. When a TOR was issued *after* subtasks, participants showed significantly less arousal than when TORs were issued while *during* subtasks. This effect was visible in two of the three evaluated GSR parameters and confirms our previous results addressing AVs [67, 68], but also other settings [3].

Regarding TOR performance, we mostly could not find significant effects except for RTd, which was lower when an subtask was interrupted. While this could be interpreted as an argument against AUIs, other TOR reaction and driving performance indicators could not confirm this. Considering our study sample, participants showed no difference in RTE and TTCoLC, and thus performed the evasive maneuver similar fast in both conditions. As non-interrupting TORs further resulted in a somewhat lower brake pedal actuation (with a mentionable effect size, however the significance was not given due to Bonferroni correction), we could interpret the results in a way, that issuing TOR after subtasks led to less abrupt maneuvers. However, the effect of “AUI behavior” on TOR performance in imminent TOR situations must be addressed in future experiments to gain additional insights. Still, our results highlight the importance of addressing multiple reaction and performance indicators as just RTd may not reveal a complete picture – faster may not always result in better overall responses.

Considering productivity, our data format currently does not allow to draw conclusions on the effect of interruption/resumption lags on NDRT performance. However, in previous work, we have shown that NDRT performance is impaired after task resumption [68]. We will further investigate the impact of TORs on task resumption in future work, including experiments addressing productivity in the “mobile office”.

Limitations

Fidelity of participants’ experiences was limited due to the somewhat artificial reading/text comprehension task, where the randomness of sentences’ content may not accurately replicate what users would experience when continuously reading and interpreting a longer piece of coherent text, as well as an artificial HUD, which does not suffer from sun glare or similar effects. Additionally, we utilized a driving simulator instead of a real SAE level 3 vehicle. On-road tests at the chosen speeds and the accompanying safety critically were not yet possible to implement at the time of executing this study. Still, we believe that the advantage of the utilized setting is to have a replicable experiment in a controlled

environment, that also does not risk participants’ safety, despite limited fidelity.

Future Work

In upcoming studies, we want to investigate more diverse driving scenarios and the important factor of motion sickness. Besides that, we intend to explore other and combined task types (such as spreadsheet manipulations, or preparation of presentation slides). This way, we further investigate the issue of interfaces fostering sequential versus parallel multitasking and extend the possibilities for office work in AVs by introducing novel interaction concepts. Based on these empirical evaluations, we can inform both industry and research on how to implement the “mobile office automated vehicle” that is attractive, effective, and safe.

7 CONCLUSION

In this paper, we highlighted the need for AVs becoming “mobile offices” that support reading/text comprehension, which could become an important selling argument for both businesses and consumers. We compared two in-vehicle UI display modalities and the impact of attention-awareness for text comprehension interfaces in SAE level 3 automated driving on a highway with imminent TOR situations. For a potentially holistic evaluation, we applied method triangulation and combined behavioral (driving and secondary task performance), physiological (workload/stress measurements), and self-rating data (standardized scales). Our results have several implications for a productive work environment in SAE level 3 AVs, that is safe (SAE J3114, [17]), but also effective, efficient, and satisfactory (ISO 9214-11, [25]):

- (1) Parallel multitasking with auditory NDRT displays in SAE level 3 automated driving causes additional cognitive workload, compared to sequential execution with visual heads-up NDRT displays.
- (2) HUDs offer the advantage of being able to force gaze towards the road and additionally offer higher productivity with lower measured workload, in comparison to auditory speech displays for text comprehension.
- (3) AUIs to issue attention-aware TORs can significantly reduce stress induced on the user. This lead to later driving reactions in imminent TOR situations, but potentially also less abrupt maneuvers.

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